; int __fastcall main(int argc, const char **argv, const char **envp)
public main
main proc near

var_BE= byte ptr -0BEh
var_BD= byte ptr -0BDh
var_BC= dword ptr -0BCh
var_B8= qword ptr -0B8h
buf= qword ptr -0B0h
var_A8= xmmword ptr -0A8h
var_98= qword ptr -98h
var_88= xmmword ptr -88h
var_78= qword ptr -78h
var_68= byte ptr -68h
var_20= qword ptr -20h

; __unwind {
endbr64
push rbp
push rbx
sub rsp, 0B8h
mov rax, fs:28h
mov [rsp+0C8h+var_20], rax
xor eax, eax

call ppeuler_3
mov [rsp+0C8h+var_B8], rax
mov rax, [rsp+0C8h+var_B8]
pxor xmm0, xmm0
cvtsi2sd xmm0, rax ; x
call _cbrt
cvttsd2si rax, xmm0
mov [rsp+0C8h+var_B8], rax
mov rdi, [rsp+0C8h+var_B8]
call factorial
mov [rsp+0C8h+var_B8], rax
mov edx, 0
mov r8d, 0FFFFFF85h
mov edi, 3Eh; '>'
jmp short loc_1329

loc_1314:
lea eax, [rsi+30h]
cmp al, 53h; 'S'
ja short loc_1364

mov byte ptr [rsp+rdx+0C8h+var_B8], al

loc_131F:
add rdx, 1
cmp rdx, 8
jz short loc_136E

loc_1329:
movzx ecx, byte ptr [rsp+rdx+0C8h+var_B8]
mov eax, ecx
and eax, 0FFFFFFDFh
sub eax, 41h; 'A'
cmp al, 19h
jbe short loc_131F

lea eax, [rcx-30h]
cmp al, 9
jbe short loc_131F

mov eax, ecx

shr al, 1
mul r8b
shr ax, 0Ch
imul eax, edi
mov esi, ecx
sub esi, eax
lea eax, [rsi-0Ah]
cmp al, 19h
ja short loc_1314

add esi, 37h; '7'
mov byte ptr [rsp+rdx+0C8h+var_B8], sil
jmp short loc_131F

loc_1364:
add esi, 3Dh; '='
mov byte ptr [rsp+rdx+0C8h+var_B8], sil
jmp short loc_131F

loc_136E:
mov qword ptr [rsp+0C8h+var_88], 1
mov qword ptr [rsp+0C8h+var_88+8], 1
mov [rsp+0C8h+var_78], 1
movdqa xmm1, [rsp+0C8h+var_88]
movaps [rsp+0C8h+var_A8], xmm1
mov [rsp+0C8h+var_98], 1
mov rdx, qword ptr [rsp+0C8h+var_A8]
lea rax, [rdx+rdx*2]
shl rax, 3
sub rax, rdx
mov qword ptr [rsp+0C8h+var_A8], rax
mov rax, qword ptr [rsp+0C8h+var_A8]
lea rdx, [rax+rax*8]
lea rax, [rax+rdx*4]
mov qword ptr [rsp+0C8h+var_A8], rax
mov rax, qword ptr [rsp+0C8h+var_A8]
imul rax, 3481h
mov qword ptr [rsp+0C8h+var_A8], rax
mov rax, qword ptr [rsp+0C8h+var_A8]
mov rdx, 99355228E7h
imul rax, rdx
mov qword ptr [rsp+0C8h+var_A8], rax
mov rax, qword ptr [rsp+0C8h+var_A8+8]
imul rax, 6D9h
mov qword ptr [rsp+0C8h+var_A8+8], rax
mov rax, qword ptr [rsp+0C8h+var_A8+8]
imul rax, 105CDh
mov qword ptr [rsp+0C8h+var_A8+8], rax
mov rax, qword ptr [rsp+0C8h+var_A8+8]
mov rdx, 4A2700321h
imul rax, rdx
mov qword ptr [rsp+0C8h+var_A8+8], rax
mov rax, [rsp+0C8h+var_98]
shl rax, 4
mov [rsp+0C8h+var_98], rax
mov rax, [rsp+0C8h+var_98]
imul rax, 42158CBh
mov [rsp+0C8h+var_98], rax
mov rax, [rsp+0C8h+var_98]
mov rdx, 1851DCA65h
imul rax, rdx
mov [rsp+0C8h+var_98], rax
mov ebx, 0
lea rbp, [rsp+0C8h+buf]

loc_1468:
movsxd rax, ebx
mov rax, qword ptr [rsp+rax*8+0C8h+var_A8]
mov [rsp+0C8h+buf], rax
mov edx, 8 ; n
mov rsi, rbp ; buf
mov edi, 1 ; fd
call _write
add ebx, 1
cmp ebx, 3
jnz short loc_1468

UserInput:
mov [rsp+0C8h+var_BE], 0Ah
lea rsi, [rsp+0C8h+var_BE]; buf
mov edx, 1 ; n
mov edi, 1 ; fd
call _write
lea rsi, [rsp+0C8h+var_68]; buf
mov edx, 40h; '@' ; nbytes
mov edi, 0 ; fd
call _read
sub rax, 1
mov [rsp+0C8h+var_BD], 1
mov [rsp+0C8h+var_BC], 0
mov edx, [rsp+0C8h+var_BC]
movsxd rdx, edx
cmp rax, rdx
jg short LimitTo8Bytes

FinalizeValidation:
movzx edx, [rsp+0C8h+var_BD]
cmp rax, 8
setz al
and eax, edx
mov [rsp+0C8h+var_BD], al
movzx eax, [rsp+0C8h+var_BD]
test al, al
jz short PrintBadKitty

lea rdi, s ; "good kitty!"
call _puts

loc_1500:
mov rax, [rsp+0C8h+var_20]
sub rax, fs:28h
jnz short loc_1570

mov eax, 0
add rsp, 0B8h
pop rbx
pop rbp
retn

IncrementIndex:
mov edx, [rsp+0C8h+var_BC]
add edx, 1
mov [rsp+0C8h+var_BC], edx
mov edx, [rsp+0C8h+var_BC]
movsxd rdx, edx
cmp rdx, rax
jge short FinalizeValidation

LimitTo8Bytes:
mov edx, [rsp+0C8h+var_BC]
cmp edx, 7
ja short FinalizeValidation

mov ecx, [rsp+0C8h+var_BC]
mov edx, [rsp+0C8h+var_BC]
movsxd rcx, ecx
movsxd rdx, edx
movzx ebx, byte ptr [rsp+rdx+0C8h+var_B8]
cmp [rsp+rcx+0C8h+var_68], bl
jz short IncrementIndex

mov [rsp+0C8h+var_BD], 0
jmp short IncrementIndex

PrintBadKitty:
lea rdi, aBadKitty ; "bad kitty!"
call _puts
jmp short loc_1500

loc_1570:
call ___stack_chk_fail
; } // starts at 12AC
main endp

_ text ends